Charotar University of Science and Technology

Education Campus, Changa – 388 421.

M Sc (BIOTECHNOLOGY)
SEMESTER II EXAMINATION
BT706: GENETICS AND BIOCHEMISTRY

Date: 04.05.2011
Time: 10.00 AM to 1.00 PM **Total Marks: 70**

Instructions to the candidates:

1. Attempt all questions
2. Questions in **Part A** should be answered in the question paper itself. Please do not write your candidate ID number on the question paper.
3. Write answers for Section I and Section II of **Part B** in separate answer sheets.

PART A **Total marks: 20**

Q1. Choose the correct option and put √ mark in front of it:

1. *N*-tosyl-L-phenylalanine chloro methyl ketone (TPCK) is an example of
a) Competitive inhibitor b) Non-competitive inhibitor c) Irreversible inhibitor
d) Uncompetitive inhibitor

2. Different enzymes that catalyze same reactions are known as:
a) holoenzyme b) apoenzyme c) allosteric enzyme d) isoenzyme

3. The following is an enzyme but **not** a protein in nature.
a) RNase b) Lysozyme c) Trypsin d) Ribozyme

4. Allosteric enzymes do obey Michaelis-Menton kinetics. [True] [False]

5. The following is **not** a monosaccharide
a) Glucose b) Sucrose c) Ribose d) Glyceraldehyde

6. The following is **not** a water soluble carbohydrate
a) Starch b) Agarose c) Dextran d) Cellulose

7. Following polymer is preferred over glycans as storage fuel.
a) Lipids b) Proteins c) Vitamins d) None of the above

8. The molecular weight of a cyclic polypeptide comprising of 50 amino acid residues is
a) 5.5 kD b) 6.0 kD c) 5.39 kD d) 5.0 kD

9. The reaction which irreversibly commits sugar to glycolytic pathway is catalyzed by phosphofructokinase. [True] [False]

10. Gluconeogenesis is a process by which organisms synthesize glucose, is a reversal of glycolysis. [True] [False]

11. The β-oxidation proper process occurs in
a) Mitochondria b) Cytosol c) Intermembrane space d) none of the above

12. Fructose 1, 6-bisphosphate is an activator for pyruvate kinase. [True] [False]

13. A phenotype is either the product of gene or of environmental influences.
[True] [False]

14. A black female is test-crossed producing 6 black offsprings. The probability that a heterozygous black female would do this by chance alone is approximately
a) 50% b) 25% c) 1% d) cannot be determined from the information given

15. How many genetically different gametes can be made by an individual of genotype *AaBbccDDEe?*
a) 8 b) 10 c) 32 d) none of the above

16. From the cross *AB/ab x AB/ab* we observe *9%* of the progeny to be *ab/ab.* The estimate of the recombination percentage between these two loci is
a) 50 b) 40 c) 35 d) 20

17. During conjugation between an F- bacterium and an Hfr bacterium, the sex factor always
a) remains in the Hfr cell
b) moves into the F- cell
(c) attaches to the bacterial chromosome at the same location in all strains
(d) is found at the rear end of the linear chromosome

18. A transducing phage:
a) contains only viral DNA
b) may contain viral and bacterial DNA
c) is sensitive to DNase
d) can never transfer extrachromosomal genes

19. Gene transfer in bacteria by transformation has the following characteristics:
a) a majority of the donor genes are transferred
b) it involves a plasmid
c) it depends on phage infection of the recipient cell
d) it can be carried out using free DNA extracted from the donor

20. Why is tetrad analysis useful for studying genetics?
a) Cells within a tetrad are haploid so genotype is directly reflected in the phenotype
b) Cells within a tetrad reflect the events of meiosis.
c) Analysis of simple systems can elucidate general rules that apply more broadly.
d) All of the above.

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PART B **Total marks: 50**

Section I

Q-1 A Attempt **any one** **(3)**

i) Black coat color in Cocker Spaniels is governed by a dominant allele *B* and red coat color by its recessive allele *b:* solid pattern is governed by (he dominant allele of an independently assorting locus *S.* and sported pattern by its recessive allele ,v. A solid-black mate is mated to a solid-red female and produces a litter of 6 pups: *2* solid black. 2 solid red, I black and white, and I red and white. Determine the genotypes of the parents.

ii) In corn, rough sheath (*rs*) is recessive to smooth sheath (*Rs*), midrib absent (*mrl*) is recessive to midrib present (*Mrl*), and crinkled leaf (*cr*) is recessive to smooth leaf (*Cr*). What are the results of testcrossing a trihybrid?

iii) Discuss the extensions of Mendel's law with example

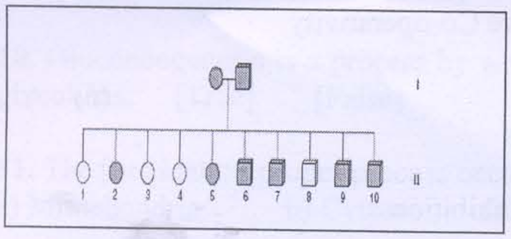
Q-1 B Attempt **any one** **(2)**

i) Black pelage of guinea pigs is a dominant trait; while is the alternative recessive trail. When a pure black guinea pig is crossed to a white one, what fraction of the black F2 is expected to be heterozygous?

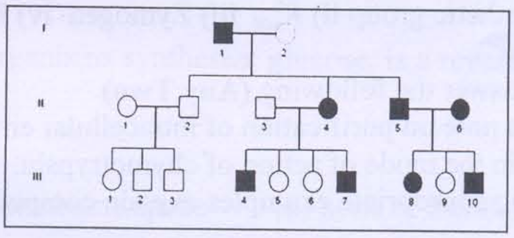
ii) In summer squash, white fruit color is governed by a dominant allele (W) and yetlow fruit color by the recessive (w). A dominant allele at another locus (S) produces disc-shaped fruit and its recessive allele (s) yields sphere-shaped fruit. If a homozygous white disc variety of genotype WVV55 is crossed with a homozygous yellow sphere variety (wws), the F1 are all white disc dihybrids of genotype *WwSs.* If the F2 |is allowed to mate at random, what would be the phenotypic ratio expected in the *f2* generation

Q-1 C Analyze the following human pedigree **(Any one)** **(3)**

i)



ii)



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Q-1 D If P and Z are two different gene of concern calculate the map distance between these genes by using the data given :PZ is 42.4% Pz is 6.9% pZ is 7.1 pz is 43.6 what would be the map distance? **(2)**

OR

Q-1 D Create a genetic map using the information provided below.
(i) A & D = 2 units (ii) B & D = 10 units (iii) C & B = 3 units (iv) C & A = 5 units

Q2. A Attempt **Any Two** of the following. **(6)**

i) Discuss the mechenisam of bacterial transformation in gram negative bacteria
ii) Explain the formation of F' with the example of arabinose locus
iii) Describe the salient features of extrachromosomal element

Q2. B Explain the following terms giving examples **(Any two)** **(4)**

i) Ordered and unordered Tetrad
ii) Hfr
iii) Specialized transduction
iv) Tra operon

Section II

Q.3. A. Describe the significance and nomenclature of fatty acids. **(4)**

OR

Q.3. A. Explain in detail the classification of carbohydrates.

Q.3. B. Write a note on **(Any Two)** **(6)**

i) Structure of keratin protein
ii) Structure of any one heteropolysaccharide
iii) Structure of spingolipids

Q.4.A Explain the steps involved in catabolism of hexoses in prokaryotes and its regulation. **(4)**

OR

Q.4.A. Explain the anapleurotic reactions and its importance.

Q.4.B. Write a note on **(Any Two)** **(6)**

i) Pyruvate dehydrogenase complex ii) Oxidation of PUFA ii) Enzyme multiplicity

Q.5.A. Derive the Michaelis-Menton equation for single substrate enzyme catalyzed reaction in presence of uncompetitive inhibitor. **(4)**

OR

Q.5.A Define the following terms: **(4)**

i) Prosthetic group **ii)** K_{cat} **iii)** Zymogen **iv)** Positive Co-operativity

Q.5.B. Answer the following **(Any Two)** **(6)**

i) Write a note on purification of intracellular enzymes
ii) Explain the mode of action of chymotrypsin.
iii) Giving appropriate examples explain competitive inhibition.